

SHESHAN S MANIYUR

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in LLM, NLP, and RAG-based systems. Proven experience designing and deploying AI applications using Python, Flask, and AWS. Expertise in building agentic workflows and managing complex context to deliver scalable, production-ready ML solutions.

SKILLS & INTERESTS

- AI & LLM Tools: LLM, RAG Systems, LangChain, LlamaIndex, OpenAI API, Prompt Engineering, Hugging Face Transformers.
- Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning.
- Programming: Python (NumPy, Pandas, Scikit-Learn), SQL.
- Data Engineering: FAISS, Chroma, Exploratory Data Analysis (EDA), Data Cleaning, Visualization.
- Cloud & DevOps: AWS (EC2, S3, VPC, Load Balancer), Flask, Git, GitHub, Linux.

RELEVANT EXPERIENCE

Self-Operating Computer (AI-Driven GUI Automation)

- Developed an **autonomous desktop-interaction agent** using multimodal vision-language models, reducing manual effort by **70%**.
- Engineered **LLM-based action planning via LangChain** workflows to convert natural-language commands into real-time GUI operations.
- Architected a model-agnostic orchestration framework compatible with **OpenAI, Llama, and Mistral** models.
- Designed a **"See Think Act"** control loop achieving <500ms latency for real-time responsiveness.
- Increased automation consistency by **25%** through optimized Python logic and dynamic tool execution.

Offline YouTube Video Summarizer

- Created a fully **offline RAG-style summarization system**, ensuring complete data privacy and zero cloud dependency.
- Integrated **Faster-Whisper ASR** achieving **95% transcription accuracy** with FAISS-based embedding retrieval.
- Developed an abstractive summarization engine using **transformer-based models** and refined prompt templates.
- Deployed a **Flask interface** for real-time summary generation, increasing content consumption speed by **5x**.

Air Quality Prediction System using Big Data Analytics

- Built an **AI pipeline** predicting air quality using real-time **Open-Meteo API data**.
- Trained and optimized models (**XGBoost, Random Forest, Logistic Regression**) achieving **99% accuracy**.
- Enhanced system performance by **35%** through advanced preprocessing and hyperparameter tuning.
- Improved performance by 35% via advanced preprocessing and hyperparameter tuning.
- Delivered visual analytics (heatmaps, confusion matrices) improving interpretability by **50%**.

LICENSES & CERTIFICATIONS

- The Complete 2024 DevOps and AWS bootcamp.

EDUCATION

JSS Science and Technology University
Masters in Data Science

Mysuru, India
2024-present

Bangalore Institute of Technology
Bachelors in Electronics and Communication Engineering (CGPA - 7.3)

Bengaluru, India
2018 – 2022

Sri Chaitanya PU College
Physics, Chemistry, Mathematics, Biology (Percentage – 83.05%)

Bengaluru, India
2016 – 2018